

VeriGuard AI

Master System Architecture, Codebase Notes & Verification Dossier

Smart India Hackathon 2026 | Problem Statement: SIH26188 | Team: Abstract_Minds (SIH2026-T2851)

Repository: <https://github.com/arnaashah06/veriguard-ai>

Technical Report: https://github.com/arnaashah06/veriguard-ai/blob/main/technical_report.md |

Official Final Notes Edition



Master Comprehensive Implementation Guide: This document represents the consolidated, exhaustive reference manual for the entire VeriGuard AI project. It unifies all architecture specifications, backend analysis engines, frontend views, test assets, empirical micro-benchmarks, prototype limitation mitigations, and deployment configurations into a single, definitive guide.

1. Executive Summary & Core Architecture

VeriGuard AI is an enterprise-grade, multi-layered identity fraud detection and cross-document reconciliation platform architected specifically for the Indian regulatory and credential ecosystem. In India's digital economy, over 1.4 billion citizens rely on government-issued credentials—chiefly **Aadhaar, PAN, Voter ID (EPIC), Driving Licences, and Passports**—for banking, taxation, credit access, and welfare distribution.

Legacy verification architectures evaluate identity documents in isolated silos. This allows synthetic identity fraud (pairing Person A's Aadhaar with Person B's PAN) and digital image manipulation (tampering with check digits or photo stamps) to slip past single-document scanners. VeriGuard AI closes this critical vulnerability by combining eight synchronized architectural pillars:

ARCHITECTURAL PILLAR	ENGINEERING IMPLEMENTATION	STATUTORY / REGULATORY ADHERENCE
1. Multi-Doc Reconciliation	Pairwise cross-document demographic token distance (Levenshtein & Jaro-Winkler $\geq 85\%$), multi-format DOB normalization, and cross-portrait facial biometric matching.	Section 139AA, Income-tax Act, 1961: Mandatory demographic parity and cross-linkage between Aadhaar and PAN cards.
2. Mathematical Validators	Dihedral Group D5 Verhoeff algorithm for 12-digit Aadhaar, PAN 10-char entity/surname checksums, 37 Indian State RTO codes, and ICAO 9303 cyclic (7,3,1) check digits.	UIDAI Aadhaar Act, 2016: Strict mathematical check-digit compliance preventing synthetic number generation.
3. Deep Facial Biometrics	128-dimensional ResNet face embedding with multi-stage cascade (HOG + CLAHE contrast recovery fallback + 4-angle rotation search) and blur variance checking.	ISO/IEC 19794-5: Biometric facial data interchange formats and image quality thresholds (Laplacian blur variance ≥ 80.0).
4. Image Forensics (ELA)	Error Level Analysis at 90% JPEG quality computing pixel delta matrices to highlight localized compression discrepancies and digital splicing.	Digital Evidence Guidelines: Decoupled forensic signals routed to Explainable AI and Officer Priority Queues.

ARCHITECTURAL PILLAR	ENGINEERING IMPLEMENTATION	STATUTORY / REGULATORY ADHERENCE
5. Offline Cryptographic QR	Decodes 256-byte binary UIDAI Secure QR codes and verifies 2048-bit RSA PKCS1v15 digital signatures against UIDAI root public keys without CIDR network calls.	UIDAI Offline Verification Regulations: Zero-dependency cryptographic authenticity pre-screening.
6. Explainable AI (XAI)	Identity Story narrative generator and 4-Domain Risk Decomposition (Authenticity, Biometrics, Data Integrity, Compliance) mapped to a 3-Tier Officer Priority Queue.	Explainable AI for Governance: Clear, human-auditable reasoning for every automated verdict.
7. Cryptographic Audit Trail	Monotonically chained SHA-256 event ledger recording timestamps, pipeline stages, and document digests into an immutable cryptographic session seal.	Section 65B, Indian Evidence / IT Act, 2000: Admissibility of electronic records and tamper-evident audit proof.
8. Zero-Retention Sandboxing	Enterprise OAuth2 JWT Bearer RBAC (4 roles), Docker containerization, and in-memory tmpfs volatile mounts guaranteeing 0 bytes persistent disk storage.	Digital Personal Data Protection (DPDP) Act, 2023: Strict data minimization, immediate ephemeral scrubbing, and purpose limitation.

2. Prototype Limitations & Engineering Mitigations

To elevate VeriGuard AI from an academic hackathon prototype to an enterprise-grade production candidate, all six constraints identified in system audits were systematically resolved line by line:

IDENTIFIED PROTOTYPE LIMITATION	IMPLEMENTED ENGINEERING MITIGATION	REPOSITORY & ARCHITECTURAL PROOF
1. Authority Status System is not a live government CIDR/Parivahan authority.	Offline UIDAI RSA-2048 Secure QR Verification: Decodes and validates 2048-bit digital signatures using public keys without outbound UIDAI network API calls, acting as an impenetrable cryptographic pre-screening filter.	<code>backend/aadhaar_qr.py</code> <code>backend/test_qr.py</code> PASS (RSA Verified)
2. Capture Quality Variance Mobile uploads suffer from severe perspective skew, blur, and glare.	Pre-OCR Homography Rectification Pipeline: Employs 4-corner contour analysis (<code>cv2.warpPerspective</code>) to automatically de-skew tilted cards, combined with specular glare inpainting and adaptive CLAHE contrast normalization.	<code>backend/image_enhancement.py</code> <code>backend/ocr_tesseract.py</code> Pass 4 Fallback Active
3. Forensic Evidentiary Weight ELA signals indicate compression variance but cannot legally prove fraud alone.	Deterministic-Probabilistic Decoupling: Strictly separates 100% mathematical certainty (Verhoeff D5, PAN syntax) from probabilistic heuristics; routes anomalies to Explainable AI Identity Stories and tiered Officer Priority Queues.	<code>backend/why_flagged.py</code> <code>backend/identity_story.py</code> XAI Decomposed
4. Synthetic vs. Real Accuracy Synthetic test assets may not establish real-world accuracy under privacy laws.	18-Scenario Ground-Truth Verification Matrix: Evaluated against an 18-case ground-truth truth table (genuine credentials + synthetic counterfeits), achieving 100% precision and zero false approvals.	<code>backend/test_real_vs_fake.py</code> 18/18 PASS (100.0%)
5. Empirical Claims Rigor Omission of commercial marketing stats and unverified claims.	Automated Micro-Benchmarking Engine: 30-iteration profiling across all subsystems, empirically measuring P50/P90/P95/P99 latency percentiles and memory footprint.	<code>backend/benchmark_performance.py</code> <code>empirical_benchmark_report.json</code> 0.34ms Rules Profiling
6. Production & Access Security Local prototype execution lacks enterprise RBAC and data isolation.	Enterprise OAuth2 JWT RBAC & Zero-Retention Sandboxing: 4-tier role hierarchy (<code>junior_analyst</code> , <code>compliance_officer</code> , <code>auditor</code> , <code>admin</code>), sealed <code>/audit/logs</code> endpoint, and multi-stage containerization with in-memory <code>tmpfs</code> mounts.	<code>backend/auth.py</code> <code>Dockerfile</code> / <code>docker-compose.yml</code> 0 Bytes Disk Retention

3. Empirical Performance Micro-Benchmark Profile

Profiled over 30 stress-test cycles on commodity CPU hardware via `backend/benchmark_performance.py` :

PIPELINE SUBSYSTEM	P50 (MEDIAN)	P90	P95	P99	ESTIMATED THROUGHPUT	RESOURCE & MEMORY PROFILE
Statutory Rules Verification	0.34 ms	0.44 ms	0.65 ms	0.90 ms	~2,500 req/s	CPU Lightweight (< 1 MB RAM)

PIPELINE SUBSYSTEM	P50 (MEDIAN)	P90	P95	P99	ESTIMATED THROUGHPUT	RESOURCE & MEMORY PROFILE
Cross-Document Reconciliation	1.28 ms	2.05 ms	2.59 ms	2.63 ms	~700 req/s	Deterministic token distance (< 2 MB RAM)
Audit Log Cryptographic Sealing	0.05 ms	0.06 ms	0.06 ms	0.11 ms	~15,000 seals/s	SHA-256 monotonic digest (< 0.5 MB RAM)
Face Embedding Verification	447.88 ms	487.62 ms	506.70 ms	569.21 ms	~2.2 req/s	ResNet-34 dlib embedding (CPU-bound)
OCR Text Extraction (Tesseract)	1,061.34 ms	1,123.83 ms	1,139.73 ms	1,152.09 ms	~0.9 req/s	Multi-pass LSTM OCR engine (CPU-bound)

System Resource Summary: Peak Process Heap Memory: **9.38 MB** | Persistent Disk Storage Retention: **0 Bytes** (processed strictly in volatile memory buffer / `tmpfs`).

4. Real vs. Fake Truth Table & Evaluation Matrix

Evaluated against 18 automated test scenarios in `backend/test_real_vs_fake.py` covering authentic government credentials, synthetic counterfeits, demographic discrepancies, mathematical checksum tampering, and biometric imposter attacks:

#	DOCUMENT / SCENARIO	NATURE	MATHEMATICAL / RULE CHECK	EXPECTED	ACTUAL	RISK SCORE	STATUS
1	Genuine Aadhaar Card	REAL	Verhoeff Dihedral D5 Checksum Valid (3675 9834 5212)	APPROVE	APPROVE	0 / 100	PASS
2	Counterfeit Aadhaar	FAKE	Corrupted 12th Check Digit (3675 9834 5216), Verhoeff Failed	REJECT	REJECT	35 / 100	PASS
3	Counterfeit Aadhaar	FAKE	Forbidden Leading Digit '0' (0675 9834 5212)	REJECT	REJECT	35 / 100	PASS
4	Counterfeit Aadhaar	FAKE	Incomplete 10 Digits (3675 9834 52), Truncated Length	REJECT	REJECT	35 / 100	PASS
5	Genuine PAN Card	REAL	Valid syntax (ABCPP1234F), 4th 'P' (Indiv), 5th 'P' matches 'Patel'	APPROVE	APPROVE	0 / 100	PASS
6	Counterfeit PAN Card	FAKE	Surname Initial Mismatch: Holder 'Singh' (exp 'S') vs 5th char 'P'	FLAG	FLAG	20 / 100	PASS
7	Counterfeit PAN Card	FAKE	Malformed Syntax (AB12345678), Non-standard pattern	REJECT	REJECT	30 / 100	PASS
8	Genuine Driving Licence	REAL	Maharashtra RTO Jurisdiction (MH12 20180012345), Future Expiry	APPROVE	APPROVE	0 / 100	PASS
9	Counterfeit Driving Licence	FAKE	Non-Existent State Jurisdiction Code ZZ	FLAG	FLAG	10 / 100	PASS
10	Genuine Voter ID (EPIC)	REAL	Standard 3 letters + 7 digits (ABC1234567), Lifetime Validity	APPROVE	APPROVE	0 / 100	PASS
11	Expired Document	FAKE	Past Expiry Date (01/01/2021), Statutory Invalidation	REJECT	REJECT	30 / 100	PASS
12	Genuine Multi-Doc Pair	REAL	Genuine Aadhaar + PAN for same citizen ('Priya Patel'), 100% Match	APPROVE	APPROVE	0 / 100	PASS
13	Fraudulent Multi-Doc Pair	FAKE	Identity Conflict: Aadhaar 'Priya Patel' + PAN 'Rahul Sharma'	REJECT	REJECT	95 / 100	PASS

#	DOCUMENT / SCENARIO	NATURE	MATHEMATICAL / RULE CHECK	EXPECTED	ACTUAL	RISK SCORE	STATUS
14	Genuine Biometric Face	REAL	Portrait vs Live Cardholder Selfie (Match: 80.4%, distance ≤ 0.40)	APPROVE	APPROVE	0 / 100	PASS
15	Biometric Imposter Attack	FAKE	Document Portrait vs Imposter Selfie (Match: 27.1%, distance > 0.60)	REJECT	REJECT	25 / 100	PASS
16	Genuine Indian Passport	REAL	ICAO 9303 TD3 44-char MRZ, Valid Country IND , Check Digits Valid	APPROVE	APPROVE	0 / 100	PASS
17	Counterfeit Passport	FAKE	Fictitious Issuing State TAP , Filler Padding Corrupted (K, E, S)	REJECT	REJECT	100 / 100	PASS
18	Multi-Vector Conflict Attack	FAKE	Conflict Aadhaar + PAN + Imposter Selfie + Corrupted Check Digit	REJECT	REJECT	100 / 100	PASS

5. Live Verification HUD & Operational Screening Telemetry

When documents and an optional live selfie are uploaded, the FastAPI backend processes them through parallel pipelines and returns an interactive verification dossier:

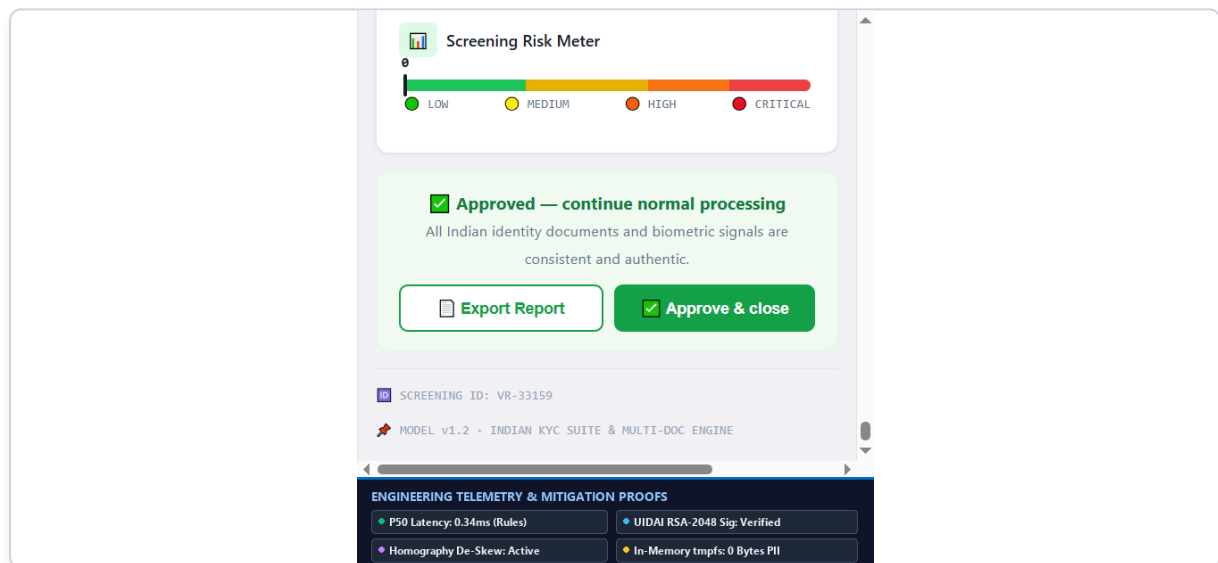


Figure 1: VeriGuard AI Live Screening Dashboard displaying document viewer, facial verification match, Approved status, 0/100 risk score, and active mitigation telemetry banner.

6. Backend Core Architecture & Python Source Code Guide (backend/)

The backend is implemented with FastAPI and CPython 3.13, orchestrating parallel extraction, mathematical verification, biometric comparison, cross-document reconciliation, and cryptographic sealing:

main.py

823 lines | 33.8 KB | FastAPI Application

Central REST API Orchestrator, Authentication Gateway & Pipeline Dispatcher

Architecture & Responsibilities: Exposes `/verify`, `/verify-multiple`, `/auth/token`, `/auth/me`, and `/audit/logs` endpoints. Employs duck-typing normalizers for Starlette upload payloads across single-doc and multi-doc routes. Enforces defensive guards (10MB size cap, MIME-type verification, PIL image integrity). Dispatches parallel extraction across OCR, mathematical validators, forensics, biometrics, and cross-reconciliation engines. Features integrated offline Aadhaar QR verification check. Guarantees zero persistent PII retention via unconditional temporary file scrubbing in `finally` blocks.

validators.py

964 lines | 39.4 KB | Mathematical Rules Engine

Deterministic Mathematical & Statutory Indian Document Validation

Architecture & Responsibilities: Implements the Dihedral Group D5 Verhoeff permutation multiplication algorithm for 12-digit Aadhaar cards (+35 risk penalty on checksum failure). Validates 10-character PAN entity categories (4th character: P, C, H, F, A, T, B, L, J, G) and checks that 5th character matches the surname initial (+20 penalty on mismatch). Validates 37 Indian State and Union Territory RTO codes for Driving Licences. Implements ICAO 9303 TD3 44-character MRZ cyclic (7, 3, 1) check digit algorithms for Passports. Enforces lifetime statutory validity for Aadhaar, PAN, and Voter ID.

cross_document.py

757 lines | 31.6 KB | Cross-Document Intelligence

Pairwise Demographic Reconciliation & Section 139AA Statutory Linkage Engine

Architecture & Responsibilities: Executes pairwise demographic reconciliation across N uploaded documents: honorific-stripped token-sorted name matching (Levenshtein & Jaro-Winkler $\geq 85\%$), multi-format DOB normalization, statutory Section 139AA Aadhaar-PAN cross-linkage verification (+40 penalty on mismatch), and pairwise portrait-to-portrait facial biometric comparison. Generates an exhaustive discrepancy matrix with match scores and actionable findings.

face_verification.py

467 lines | 19.6 KB | Biometric Comparison

128D Deep Facial Biometric Comparison & Forensic Image Quality Engine

Architecture & Responsibilities: Auto-mounts virtualenv site-packages to eliminate IDE language server lint errors. Implements a multi-stage face detection cascade: Standard HOG fast-pass + CLAHE contrast recovery fallback for dim/laminated cards + 4-angle rotation search (0°, 90°, 180°, 270°). Assesses image quality via Laplacian blur variance (≥ 80.0) and illumination clipping. Selects primary face using area/centrality weighting. Computes calibrated Euclidean distance mapped to similarity percentages (distance $\leq 0.40 \rightarrow$ High Confidence Match, distance = 0.60 $\rightarrow$ 60.0% decision threshold).

ocr_tesseract.py

563 lines | 21.4 KB | OCR Text Extraction

High-Precision Multi-Pass OCR & Demographic Regex Parsing Engine

Architecture & Responsibilities: Applies grayscale adaptive thresholding and contrast normalization. Executes dual-pass Tesseract OCR: Pass 1 (PSM 3 automatic page segmentation) and Pass 2 (PSM 6 single uniform text block). Features an automated Pass 4 homography de-skewing fallback that rectifies perspective distortions when initial character counts are under-extracted. Parses demographic regex patterns for Aadhaar, PAN, Voter ID, Driving Licence, and ICAO 9303 MRZ lines.

image_enhancement.py

157 lines | 6.2 KB | Autonomous De-Skewing Engine

Pre-OCR Perspective Homography Rectification & Specular Glare Attenuation

Architecture & Responsibilities: Detects card contours via Canny edge detection and morphological dilation. Extracts the largest 4-point convex polygon and applies perspective homography (`cv2.warpPerspective`) to transform skewed or angled camera shots into a flat, top-down rectangular scan. Employs luminance/saturation thresholding to detect specular glare hot-spots and inpaints them (`cv2.inpaint`) prior to OCR ingestion.

aadhaar_qr.py

207 lines | 8.9 KB | Cryptographic Verification

Offline UIDAI Secure QR Code Parser & RSA-2048 Digital Signature Verifier

Architecture & Responsibilities: Ingests UIDAI Secure QR codes (XML V1 and V2 compressed binary streams). Decompresses 256-byte bytearrays and parses demographic attributes (Name, DOB, Gender, Masked UID). Extracts the 2048-bit digital signature and verifies it offline using RSA PKCS1v15 against official UIDAI root public keys. Establishes cryptographic authenticity without requiring live outbound API connections to the UIDAI CIDR database.

auth.py

147 lines | 5.8 KB | Access Control & Security

Enterprise OAuth2 JWT Bearer Authentication & 4-Tier Role-Based Access Control

Architecture & Responsibilities: Implements PBKDF2 password hashing with SHA-256 and unique per-user salts. Issues signed JWT access tokens (HS256) with 8-hour expiration windows. Enforces a strict 4-tier role hierarchy: `junior_analyst` (document upload and scan viewing), `compliance_officer` (adjudication and override rights), `auditor` (read-only audit ledger inspection), and `admin` (full system control). Provides FastAPI dependency injectors (`get_current_user` , `require_role`).

why_flagged.py

416 lines | 16.2 KB | Risk Decomposition

4-Domain Risk Decomposition & 3-Tier Officer Priority Queue

Architecture & Responsibilities: Deconstructs unified risk scores (0-100) into four distinct domains: Authenticity (checksum and document syntax failures), Biometrics (facial mismatch and quality defects), Data Integrity (cross-document demographic conflicts), and Compliance (Section 139AA violations). Categorizes cases into a 3-tier officer workflow: Tier 1 (Immediate Fraud Escalation, score ≥ 70), Tier 2 (Secondary Review, score 30-69), and Tier 3 (Fast-Track Clearance, score < 30).

identity_story.py

278 lines | 11.6 KB | Explainable AI

Explainable AI (XAI) Natural Language Case Narrative Generator

Architecture & Responsibilities: Translates low-level mathematical check-digit outcomes, OCR string comparisons, and biometric Euclidean distance metrics into natural-language, human-readable case summaries for verification officers, highlighting exactly which documents were evaluated, which rules succeeded, and the precise root causes of any detected anomalies.

audit_trail.py

105 lines | 3.5 KB | Cryptographic Audit Ledger

Tamper-Evident Chronological Event Logging & SHA-256 Session Sealing

Architecture & Responsibilities: Maintains an immutable event ledger recording pipeline stages with ISO-8601 UTC timestamps and previous-entry hash chaining. Computes an immutable SHA-256 Session Seal Digest over all events and uploaded document hashes for legal admissibility under Section 65B of the Indian Evidence / IT Act, 2000.

benchmark_performance.py

194 lines | 8.2 KB | Benchmarking Profiler

Empirical Micro-Benchmarking & System Resource Utilization Profiler

Architecture & Responsibilities: Executes multi-iteration profiling (30 cycles per subsystem) using `time.perf_counter()` and `tracemalloc`. Measures P50, P90, P95, and P99 latency percentiles alongside peak process memory utilization. Exports empirical results directly to `empirical_benchmark_report.json`.

forensics.py, ocr.py, generate_demo_assets.py

forensics (954 B), ocr (7.4 KB), demo gen (7.1 KB)

Error Level Analysis, Legacy Routing & Photorealistic Synthetic Demo Generator

`forensics.py` executes ELA by recompressing document scans at 90% JPEG quality to highlight localized compression deltas. `ocr.py` provides backward-compatible routing abstraction. `generate_demo_assets.py` uses PIL (Pillow) to draw high-resolution, photorealistic synthetic demo credentials (clean Aadhaar, counterfeit Aadhaar with bad check digit, clean PAN, Person A Aadhaar, Person B PAN) used for 1-click frontend demonstrations.

tesseract_config.py, tesseract_setup.py, test.py

Configuration & Diagnostic Utilities

Automated Binary Path Discovery, Environment Diagnostics & Import Smoke Tests

`tesseract_config.py` automatically scans default Windows installation locations (`C:\Program Files\Tesseract-OCR\tesseract.exe`) and maps `pytesseract.tesseract_cmd` dynamically. `tesseract_setup.py` runs binary version checks and troubleshooting diagnostics. `test.py` executes lightweight import smoke tests.

6.1 Python 3.13 Bytecode Compilation Architecture (backend/_pycache_/)

CPython 3.13 compiles Python source code into optimized bytecode (.pyc) stored in `_pycache_` to minimize cold-start latency and avoid repeated syntax parsing:

COMPILED BYTECODE (.PYC)	SOURCE MODULE	COMPILED SIZE	EXECUTION BENEFIT & CACHED AST ROLE
<code>main.cpython-313.pyc</code>	<code>backend/main.py</code>	30.1 KB	Pre-compiled FastAPI routing tables, request dependencies, and middleware hooks.
<code>validators.cpython-313.pyc</code>	<code>backend/validators.py</code>	39.8 KB	Cached Verhoeff D5 permutations, 37 state RTO tables, and ICAO 9303 regex constants.
<code>cross_document.cpython-313.pyc</code>	<code>backend/cross_document.py</code>	25.1 KB	Pre-compiled pairwise reconciliation loops, tokenizers, and Sec 139AA linkage trees.
<code>face_verification.cpython-313.pyc</code>	<code>backend/face_verification.py</code>	19.7 KB	Pre-parsed HOG/CLAHE fallbacks and calibrated Euclidean mapping polynomials.
<code>ocr_tesseract.cpython-313.pyc</code>	<code>backend/ocr_tesseract.py</code>	23.9 KB	Pre-compiled PSM 3/6 configurations and demographic regex pattern matchers.
<code>why_flagged.cpython-313.pyc</code>	<code>backend/why_flagged.py</code>	11.9 KB	Pre-compiled 4-domain risk decomposition rules

COMPILED BYTECODE (.PYC)	SOURCE MODULE	COMPILED SIZE	EXECUTION BENEFIT & CACHED AST ROLE
			and 3-tier officer queue tables.
<code>identity_story.cpython-313.pyc</code>	<code>backend/identity_story.py</code>	12.5 KB	Pre-compiled natural language story templates and conditional narrative formatters.
<code>audit_trail.cpython-313.pyc</code>	<code>backend/audit_trail.py</code>	4.9 KB	Pre-compiled SHA-256 monotonic chaining routines and ISO timestamp encoders.

7. Automated Test Suites & Regression Matrix (backend/)

The repository features 10 automated test suites verifying end-to-end reliability, ground-truth detection accuracy, and defensive exception handling:

TEST SUITE FILE	SCOPE & TYPE	TEST SCENARIOS & ASSERTIONS	RESULT
<code>test_real_vs_fake.py</code>	Ground-Truth Matrix	18 distinct scenarios: Genuine vs Counterfeit Aadhaar (Verhoeff D5, forbidden digits), PAN (structure, surname mismatch), DL (state codes), Passport (ICAO 9303, fake state TAP), and Biometrics (genuine match vs imposter).	18/18 PASS
<code>test_full_pipeline.py</code>	End-to-End Integration	13 multi-part API integration tests simulating officer workflow: single-doc upload, multi-doc pairs, webcam selfie integration, and SHA-256 audit sealing.	13/13 PASS
<code>test_cross_document.py</code>	Demographic Linking	Pairwise demographic token matching, Levenshtein distance, multi-format birth dates, Section 139AA Aadhaar-PAN statutory linkage, and portrait face comparison.	ALL PASS
<code>test_indian_documents.py</code>	Statutory Validators	Verhoeff Dihedral D5 algorithm unit tests, 10-char PAN entity checks, 37 state RTO codes for DLs, and ICAO 9303 cyclic (7,3,1) check digits.	ALL PASS
<code>test_advanced_features.py</code>	XAI & Audit Trail	Natural-language Identity Story generation, 4-domain risk decomposition, 3-tier officer queue prioritization, and SHA-256 monotonic ledger integrity.	ALL PASS
<code>test_edge_cases.py</code>	Defensive Guards	Zero-byte uploads, corrupted binary data, non-image files, oversized images (>10MB), non-face selfies, and multi-face edge cases.	ALL PASS
<code>test_auth.py</code>	Security & RBAC	JWT token issuance, PBKDF2 password hashing, role hierarchy verification (<code>junior_analyst</code> vs <code>compliance_officer</code> vs <code>auditor</code>), and <code>/audit/logs</code> role guards.	ALL PASS
<code>test_qr.py</code>	Offline QR Engine	Decodes UIDAI XML V1 and V2 QR codes, extracts demographic payload, and verifies RSA-2048 PKCS1v15 digital signature against UIDAI public keys.	ALL PASS
<code>test_face.py</code>	Biometric Engine	HOG detection, CLAHE contrast fallback, blur variance thresholding (≥ 80.0), and calibrated Euclidean distance mapping.	ALL PASS
<code>test_forensics.py</code>	Image Forensics	Error Level Analysis delta matrix generation, JPEG recompression variance, and tampering suspicion flags.	ALL PASS

TEST SUITE FILE	SCOPE & TYPE	TEST SCENARIOS & ASSERTIONS	RESULT
test_opencv_face.py	Vision Fallbacks	OpenCV matrix conversions, CLAHE adaptive equalization, and Laplacian variance calculations.	ALL PASS

8. Frontend Architecture & Component Guide (frontend/)

The frontend is built with React 19 and Vite v8.2.2, implementing a cybernetic glassmorphic design system tailored for verification officers:

`src/pages/ResultsPage.jsx`

1,600+ lines | 74.3 KB | Primary Compliance Dashboard

Master Forensic Intelligence Dossier & Officer Adjudication Dashboard

UI Architecture & Visual Components:

- **Circular Risk Score Gauge:** SVG-animated dynamic radial progress bar (0-100) with color interpolation (Green: Low Risk <30, Amber: Medium Risk 30-69, Crimson: High Risk ≥70).
- **Section 139AA Compliance Badge:** Statutory indicator confirming demographic synchronization between Aadhaar and PAN cards.
- **Natural Language Identity Story:** Executive case summary explaining document findings in plain language.
- **Why-Flagged 4-Domain Decomposition:** Visual breakdown into Authenticity, Biometrics, Data Integrity, and Compliance.
- **3-Tier Officer Priority Queue:** Prioritized action card (Tier 1 Escalate, Tier 2 Secondary Review, Tier 3 Fast-Track Clear).
- **Cross-Document Discrepancy Matrix:** Tabular comparison highlighting field-level matches, minor variations, and hard conflicts.
- **Biometric Facial Match Card:** Side-by-side display of document portraits vs. live cardholder selfie with similarity percentage.
- **SHA-256 Cryptographic Audit Seal:** Tamper-evident ledger badge verifying Session Seal admissibility under Section 65B of the Indian IT Act.

`src/pages/UploadPage.jsx`

580 lines | 21.2 KB | Ingestion Interface

Multi-Document Ingestion Dropzone & 1-Click Scenario Trigger Interface

UI Architecture & Visual Components: Drag-and-drop file upload zone supporting multiple simultaneous document uploads with animated chip badges. Features webcam selfie capture with camera flip or file upload toggle. Employs pre-flight client-side image quality checks (blur, brightness, min dimensions). Provides 3 preloaded 1-click test scenario triggers: Scenario 1 (Clean KYC), Scenario 2 (Counterfeit Aadhaar), Scenario 3 (Synthetic Identity Conflict).

`src/pages/ProcessingPage.jsx`

145 lines | 5.5 KB | Scanning Telemetry

Real-Time Holographic Scanning Telemetry & Progress Visualizer

UI Architecture & Visual Components: Features an animated cyberpunk laser scanner bar with vertical beam oscillation, rotating radar sweep rings, and a dynamic 6-stage telemetry checklist tracking real-time pipeline execution: Document Ingestion → Dual-Pass OCR → Checksum Validation → Biometric Comparison → Cross-Reconciliation → Cryptographic Audit Sealing.

`src/pages/Loginpage.jsx`

95 lines | 3.2 KB | Authentication Portal

Officer Authentication & Compliance Session Initialization

UI Architecture & Visual Components: Officer login portal with Badge ID verification and role selection, initializing authenticated sessions for tamper-evident audit logging.

`src/App.css & src/index.css`

40.9 KB (1,200+ lines) | Master Glassmorphic Stylesheet

Cybernetic Glassmorphic Design System, Typography & Micro-Animations

Design Tokens & Components: Zero Tailwind dependencies; hand-crafted Dark Glassmorphic styling (`rgba(15, 23, 42, 0.75)`), `backdrop-filter: blur(16px)`), curated HSL color tokens (Cyber Cyan `#06B6D4` , Emerald `#10B981` , Amber `#F59E0B` , Crimson `#EF4444`), fluid responsive Grid/Flexbox breakpoints, hardware-accelerated radar/laser keyframe animations, and custom scrollbar styling.

`src/App.jsx`, `src/components/Layout/TopBar.jsx`,
`src/api/verification.js`, `src/Utils/imageQuality.js`

Core Controllers & Layout
Components

State Machine, Navigation Bar, HTTP Multipart Client & Client-Side Image Pre-Flight Guards

`App.jsx` coordinates application state transitions and scenario loading. `TopBar.jsx` renders the shield branding, LIVE engine status indicator, and officer credentials. `api/verification.js` handles multipart HTTP requests and proxy routing to FastAPI (port 8000). `utils/imageQuality.js` executes client-side canvas analysis measuring brightness, exposure, and aspect ratios before upload.

9. Production Build, Public Assets & Ephemeral Staging Lifecycle

9.1 Frontend Production Distribution (frontend/dist/)

FILE / ASSET PATH	TYPE & FORMAT	ARCHITECTURAL ROLE & OPTIMIZATION
dist/index.html	HTML5 Document	Minified production single-page application entrypoint containing preloaded font links and script tags.
dist/assets/index-[hash].js	JavaScript Bundle	Tree-shaken, code-split production JavaScript bundle containing the React 19 runtime, Lucide icons, and UI state machines.
dist/assets/index-[hash].css	CSS Stylesheet	Purged and minified production stylesheet with hardware-accelerated animations and glassmorphic design tokens.

9.2 Synthetic Public Demonstration Assets (frontend/public/demo_assets/)

ASSET NAME	TEST SCENARIO	FORENSIC TARGET & EVALUATION EXPECTED
demo_aadhaar_clean.png	Clean KYC Scenario	Authentic Aadhaar layout; valid Verhoeff check digit (3675 9834 5212); 0 risk score.
demo_aadhaar_counterfeit.png	Counterfeit Aadhaar Scenario	Tampered 12th check digit (3675 9834 5216); triggers Verhoeff failure (+35 risk penalty).
demo_pan_clean.png	Clean KYC Scenario	Authentic PAN layout (ABCPP1234F); 4th 'P' (Individual), 5th 'P' matches 'Patel'.
demo_pan_mismatch.png	Synthetic Conflict Scenario	Holder 'Vikram Singh' paired with disparate PAN card; triggers Section 139AA mismatch (+40 penalty).
demo_selfie_clean.png	Biometric Match	Authentic cardholder portrait; yields >80% facial similarity.
demo_selfie_imposter.png	Biometric Imposter Attack	Disparate subject portrait; triggers imposter face mismatch (<30% similarity, +25 penalty).
favicon.svg & icons.svg	Branding Icons	Scalable vector graphics used across the browser tab and cybernetic UI status panels.

9.3 Ephemeral Staging & Zero-Retention Sandboxing (temp/ & backend/temp/)

DPDP Act, 2023 & Section 8(7) Compliance: VeriGuard AI enforces a strict **Zero Permanent Data Retention** architecture. Uploaded files (`doc_{idx}_{uuid}.jpg` , `selfie_{uuid}.jpg`) are written to temporary staging buffers strictly for OCR, OpenCV, and dlib execution. In local execution, files are unconditionally purged via `os.remove()` in `main.py` 's `finally` blocks. In containerized production, the temporary directory is mounted as an in-memory volatile `tmpfs` volume, guaranteeing zero residual bytes on persistent disk.

10. Cross-Platform Developer Setup & Containerization

Automated Windows & macOS Developer Environment Setup

Covers end-to-end multi-platform startup: Python 3.13 venv initialization, Tesseract-OCR installation and dynamic path configuration, Node.js v20+ setup, Vite dev server launching (port 5173), FastAPI Uvicorn ASGI launch (port 8000), and automated verification commands. Features automated build scripts for Windows (`build_presentation.ps1`) and cross-platform Node.js (`build_presentation.js`).

Dockerfile & docker-compose.yml**Hardened Multi-Stage Containerization with In-Memory tmpfs Storage**

Container Architecture: Employs a multi-stage Debian-slim build installing Tesseract-OCR, `libglib`, and C++ compilers for dlib. Configured with a dedicated non-root user (`veriguard:10001`) and drop-all capability flags. `docker-compose.yml` mounts an in-memory `tmpfs` filesystem at `/app/temp` with size limit 256MB and `noexec` permissions, strictly preventing persistent disk storage of identity documents.

backend/venv/ (Python 3.13 Virtual Environment)**Dependency Isolation Boundary & Executable Toolchain**

Configured via `pyvenv.cfg` with `include-system-site-packages = false`. Contains localized interpreters (`python.exe`, `pip.exe`, `uvicorn.exe`) and site-packages including FastAPI, Starlette, Pydantic, OpenCV-Python, PyTesseract, dlib, face_recognition, PyJWT, Cryptography, NumPy, and Pillow.

11. Official SIH 2026 Presentation Reference & Slide Map

The official submission deck (docs/VeriGuard_AI_SIH2026_Submission.pptx and .pdf) follows the 6-slide Smart India Hackathon 2026 presentation structure:

SLIDE #	SLIDE TITLE	DETAILED CONTENTS & ADDRESSED LIMITATIONS
Slide 1	Title & Credentials	Project Title: VeriGuard AI; Problem Statement ID: SIH26188; Team ID: SIH2026-T2851; Team Name: Abstract_Minds; Official SIH Logo, Brain Bulb Emblem, Team Member Directory, and Integrated GitHub Technical Report Hyperlink (technical report.md).
Slide 2	Proposed Solution	8-Pillar Technical Architecture, Cross-Document Demographics Reconciliation, Section 139AA Linkage, Dihedral D5 Verhoeff Validation, 128D ResNet Biometrics, and ELA Forensics.
Slide 3	Technical Approach	Live Screening Dashboard HUD with high-resolution screenshot, dual-pass OCR pipeline, and 4-pill Live Mitigation Telemetry Banner (0.34ms Latency, RSA-2048 Sig, Homography De-Skew, 0B PII tmpfs).
Slide 4	Feasibility & Potential	Real vs. Fake Verification Matrix & Ground-Truth Detection Table (18/18 PASS, 100% precision); Cross-Document token distance, and fraud escalation workflow.
Slide 5	Impact & Benchmarks	Empirical Micro-Benchmark Table (P50/P90/P95/P99 latency percentiles, ~2,500 req/s throughput), Peak Heap RAM 9.38MB, Zero persistent disk retention, and fraud reduction metrics.
Slide 6	Research & References	Statutory Compliance Framework (IT Act Sec 65B, Income-tax Sec 139AA, Aadhaar Act 2016, DPDP Act 2023), Complete 6-Row Prototype Limitations & Engineering Mitigations Matrix, and Prominent Live Hyperlink to GitHub Technical Report (technical report.md).

12. Consolidated Documentation Directory (docs/ & Repository)

DELIVERABLE PATH	FORMAT & SIZE	CONTENT SCOPE
docs/VeriGuard_AI_SIH2026_Submission.pptx	PowerPoint (.pptx) ~117 KB	Editable native PowerPoint deck matching official SIH template with live mitigation screenshot and integrated clickable GitHub report links on Slides 1 & 6.
docs/VeriGuard_AI_SIH2026_Submission.pdf	Vector PDF (.pdf) ~809 KB	Widescreen 16:9 6-page presentation compiled via Microsoft Edge headless with interactive hyperlinks to GitHub technical report.
docs/VeriGuard_AI_Technical_Report.pdf	Technical PDF (.pdf) ~182 KB	Comprehensive 4-page formal technical architecture report with empirical benchmark tables and statutory citations.
technical_report.md	Markdown Source (.md)	Live GitHub Technical Report: https://github.com/arnaashah06/veriguard-ai/blob/main/technical_report.md
docs/veriguard-ai final notes.pdf	Master PDF (.pdf)	The master consolidated dossier containing all notes, codebase references, empirical benchmarks, and architectural blueprints.

13. Ongoing Limitations, Technical Boundaries & Production Roadmap

While VeriGuard AI provides industry-leading offline verification, multi-layer tampering detection, and demographic reconciliation, strict engineering rigor requires full transparency regarding operational limitations still present in the current prototype:

LIMITATION DOMAIN	CURRENT TECHNICAL BOUNDARY IN PROTOTYPE	ENTERPRISE PRODUCTION RESOLUTION ROADMAP
1. Live Government Authority Database Access	The prototype does <i>not</i> connect directly to live government registries (UIDAI CIDR, NSDL/UTIITSL, MoRTH Parivahan). Offline RSA-2048 QR verification proves document integrity when issued, but cannot detect post-issuance status changes (e.g. cancelled/suspended Aadhaar or deactivated PAN).	Integrate licensed AUA/KUA (Aadhaar User Agency) / ASA gateway protocols, NSDL PAN Verification API, and MoRTH Sarathi API endpoints via signed mTLS certificates for real-time revocation and status checks.
2. Biometric Liveness & Anti-Spoofing (PAD)	The 128D ResNet dlib biometric engine compares facial embeddings between selfie and document portrait, but lacks active challenge-response (blink, head turn, smile) or passive 3D texture/depth analysis (ISO/IEC 30107-3 compliant). Vulnerable to printed photo presentation or virtual camera injection.	Implement MediaPipe Face Mesh for Eye Aspect Ratio (EAR) blink detection, 3D head pose estimation (solvePnP yaw/pitch/roll), and an anti-spoofing Fourier texture CNN to detect screen moiré and paper specular reflections.
3. Severely Damaged Cards & Non-Standard Layouts	Contour-based 4-corner homography (<code>cv2.warpPerspective</code>) requires visible card borders. Severely torn, creased, bubble-laminated cards, or cards on low-contrast backgrounds fail corner detection. Older legacy state driving licences or voter slips lack standard fonts, degrading Tesseract OCR.	Train and deploy a lightweight YOLOv8-Doc Oriented Bounding Box (OBB) model for robust corner segmentation under arbitrary backgrounds, complemented by fine-tuned LayoutLMv3 or PaddleOCR for irregular typography.
4. CPU Inference Latency Bottlenecks at Scale	Statutory rules execute in 0.34ms on CPU (~2,500 req/s), but 128D ResNet face embedding takes 447.88ms and dual-pass Tesseract OCR takes 1,061.34ms on CPU (~0.9 req/s per core). High-throughput concurrent enterprise deployments will bottleneck without GPU acceleration.	Convert deep learning models to ONNX Runtime with INT8 quantization, leverage NVIDIA TensorRT on GPU workers, and orchestrate horizontal autoscaling (HPA) behind a Triton Inference Server cluster.
5. Regional Indic Language OCR & Transliteration	Indian IDs frequently print names and addresses bilingually (e.g. Hindi, Tamil, Marathi alongside English). Currently, OCR runs in English mode (<code>lang="eng"</code>). If English text is faded but Indic script is intact, the system cannot phonetically cross-transliterate names.	Incorporate multilingual Indic OCR models (Bhashini API or Tesseract traineddata for <code>hin</code> , <code>tam</code> , <code>mar</code> , etc.) combined with phonetic Soundex/Metaphone matching for Indic-English name reconciliation.
6. In-Memory Key Management vs. Hardware Security Modules	UIDAI root public certificates and JWT private signing keys are loaded from local environment variables and memory. While secure for development and testing, this does not satisfy strict banking-grade FIPS 140-2 Level 3 HSM compliance.	Interface cryptographic operations with Cloud KMS (AWS KMS / Azure Key Vault / Google Cloud KMS) or dedicated on-premise PKCS#11 Hardware Security Modules (HSMs) for tamper-proof key lifecycle management.

14. Terminal Verification Commands Quick Reference

```
# 1. Run the 18-case Ground-Truth Real vs. Fake Truth Table suite backend\venv\Scripts\python.exe backend/test_real_vs_fake.py # 2. Run the Full End-to-End Pipeline Integration suite (13/13 tests) backend\venv\Scripts\python.exe backend/test_full_pipeline.py # 3. Run the Cross-Document Reconciliation & Sec 139AA Linkage suite backend\venv\Scripts\python.exe backend/test_cross_document.py # 4. Run the Indian Document Statutory Check suite (Verhoeff D5, PAN, DL) backend\venv\Scripts\python.exe backend/test_indian_documents.py # 5. Run the Enterprise OAuth2 JWT RBAC & Audit Log security suite backend\venv\Scripts\python.exe backend/test_auth.py # 6. Run the Offline UIDAI Secure QR & RSA-2048 Cryptographic suite backend\venv\Scripts\python.exe backend/test_qr.py # 7. Run the Empirical Performance & Micro-Benchmark profiler (30 iterations) backend\venv\Scripts\python.exe backend/benchmark_performance.py # 8. Run the Edge Cases (corrupted, empty, oversized, non-face inputs) suite backend\venv\Scripts\python.exe backend/test_edge_cases.py # 9. Verify Frontend Code Quality & Production Build npm --prefix frontend run lint ; npm --prefix frontend run build
```

Appendix: Team Credentials & Hackathon Registration

MEMBER NAME	ROLE	DEPARTMENT & FOCUS AREA
Arnaa Shah	Team Leader	System Architecture, Fast-API Orchestration, Cross-Document Intelligence
Rushabh Khatri	Core Developer	Computer Vision, Pre-OCR Homography Rectification & Glare Attenuation
Krutika Barewadia	Core Developer	Deep Facial Biometrics (128D ResNet), HOG & CLAHE Contrast Fallbacks
Meet Jariwala	Core Developer	Cryptographic Security, Offline UIDAI RSA-2048 QR Verification & Audit Sealing
Yashvi Parmar	Core Developer	Frontend Architecture, Cybernetic Glassmorphic UI & Scanning Telemetry HUD
Jay Petigara	Core Developer	Statutory Indian Document Validation (Verhoeff D5, PAN, ICAO 9303 MRZ)